

An Experimental Investigation of the Clausemate Condition in German Multiple Sluicing

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1. Introduction

Sluicing refers to an utterance that contains a seemingly nonsentential *wh*-phrase that conveys a propositional meaning and which displays the syntactic distribution of a *wh*-clause (Ross 1969), see (1) and (2B). The propositional meaning of a sluice is retrieved by reference to an *antecedent* in the proximate discourse context. In this paper, we adopt the *sententialist* perspective of sluicing, according to which the *wh*-phrase is the only pronounced phrase in an otherwise unpronounced yet syntactically present *wh*-clause (see Merchant 2019, Lasnik & Funakoshi 2019, and references in both). Following convention, we refer to the non-pronunciation as *ellipsis* and represent it with strikethrough (3).¹

- (1) Nora painted something, but I just don't know **what**. (where the sluiced *wh*-phrase is boldfaced)
(2) A: Someone has stolen Kieran's book. B: **Who**?
(3) ... but I just don't know **what** ~~Nora painted~~. (sententialist analysis of (1))

The focus of this paper is *multiple sluicing* (MS), which are configurations displaying two (or more) sluiced *wh*-phrases in one elliptic clause (4). Reading from right to left, we henceforth refer to the sluiced *wh*-phrases in an MS configuration as *swh1* and *swh2*. We refer to the phrases in the antecedent with which *swh1* and *swh2* respectively co-refer – which are italicized in (4) – as *correlate1* and *correlate2*.²

- (4) *Everyone ate something*, but I just don't know **who** ~~ate~~ **what**.

Specifically, we focus on MS configurations in which the antecedent is biclausal, where the embedded clause is selected by the root clause verb (henceforth *complex antecedent multiple sluicing*, *caMS*). Researchers have observed that *caMS* is judged as highly degraded / unacceptable when at least two of the sluiced *wh*-phrases' correlates in the antecedent utterance are separated by a finite clause boundary (Takahashi 1994, Nishigauchi 1998, Merchant 2001, Lasnik 2014, Abels & Dayal 2022), compare (4) and (5).³ This observation is known as the *Clausemate Condition* (CC).⁴

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¹ We adopt the sententialist perspective solely for expositional ease. We believe that our results could be reframed in *nonsententialist* terms (e.g., an extended version of Nykiel & Kim 2022) without impacting the validity of our conclusions.

² We remain agnostic about the syntactic derivation of MS throughout this article. To avoid notational clutter, we represent sluiced *wh*-phrases as occupying the positions that they would occupy in their nonelliptic, multiple *wh*-question counterparts.

³ The examples in (i) and (ii), from Barrie (2007:270) and Lasnik (2014:10) respectively, show that the CC does not apply in English *to*-infinitival clauses:

- (i) A certain traveler decided where to eat in a certain city. I forget which traveler in which city.
(ii) ? Some of the students wanted to go to some of the lectures, but I'm not sure which to which.

⁴ Other similar names for this observation include the *Clausemate Restriction* (Takahashi 1994), the *Clausemate Requirement* (Nishigauchi 1998), and the *Clausemate Constraint* (Lasnik 2014).

- (5) Harriet claimed [_{CP} that *every teacher spoke with some student*], but I don't know **which teacher with which student**.
- (6) * [_{CP} *Every teacher* reported [_{CP} that Harriet spoke *with some student*]], but I don't know **which teacher with which student**.

Barros & Frank's (B&F, 2022) recent analysis of the CC assumes that *swh2* undergoes QR to the left-periphery of the root clause in the elliptic clause in caMS and proposes that QR is illicit if the phrase that undergoes QR originates in a clause containing a *shifty subject*:

- (7) **Shifty subject**: The subject of a clause C is shifty when it shifts attention away from the most salient discourse referent (in the sense of Karttunen 1976, henceforth *d-ref*) evoked in the clause that embeds C. (adapted from Barros & Frank 2022:5)

B&F interpret the observation that the caMS configuration exemplified by (8) is acceptable as support for their analysis. For them, (8) is acceptable because the CC is suspended when the embedded clause's subject *S_{emb}* is not shifty (henceforth, *stable*), e.g., when *S_{emb}* co-refers with the most salient *d-ref* in the matrix clause:

- (8) [_{CP} [_{DP} Some student]_i said [_{CP} that he_i met with some professor]], but I can't recall **which student with which professor**. (adapted from Barros & Frank 2022:5)

B&F's argument presupposes that when hearers judge (8) as acceptable, they are retrieving a biclausal elliptic clause that is syntactically isomorphic to the entire antecedent utterance (henceforth, the *full elliptic source*) (9). But assuming that elliptic clauses and their antecedents need only display syntactically parallel argument domains (Rudin 2019, Anand et al. 2022), it is plausible that hearers are actually retrieving a monoclausal elliptic clause corresponding to the embedded CP of the antecedent utterance instead (henceforth, the *short elliptic source*) (10) (Abels & Dayal 2017, 2022). If true, correlate1 in (8) is actually the embedded subject, and therefore (8) does not qualify as a CC-environment in the first place. Although B&F present acceptable examples such as (11) as unambiguous cases of genuine suspension of the CC (as, according to them, only a full elliptic source is available for (11)), the existence of such utterances does not logically preclude the possibility that configurations such as (8) are acceptable because hearer is employing a short elliptic source.

- (9) ... [_{CP} [**which student**]_i said [_{CP} that he_i met with **which professor**]]
- (10) ... [_{CP} **which student must have met with which professor**]
- (11) [_{CP} *Some student* claimed [_{CP} that there was a problem *with some professor*]], but I can't recall **which student with which professor**. (adapted from Barros & Frank 2022:10)

The difficulty in determining whether hearers may – or perhaps *must* – retrieve a short elliptic source in MS configurations such as (8) is exacerbated by the fact that no study in the previous literature reports which potential interpretation – *full source* or *short source* – hearers are judging. In cases where the possible interpretations hardly differ in meaning (e.g., (9) versus (10)), this oversight is understandable, as hearers are probably unable to determine for themselves to which interpretation their acceptability judgment corresponds (also, an acceptability judgment might reflect both interpretations simultaneously).

In short, cases such as (8) appear structurally ambiguous in English. The principal aim of the study we report here was to bring clarity to the above-described situation by conducting a formal acceptability judgment experiment on MS configurations that are structurally similar to (8) but which are not ambiguous regarding the full source or short source nature of the elliptic clause. The method we adopted for avoiding ambiguity was to shift from English to a language that morphologically case-marks its DPs (we chose German) and then ensure that the relevant co-referential DPs in the antecedent utterance (i.e., *some student* and *he* in (8)) bear distinct case-markers, for instance DAT and NOM. A full or short source interpretation is then forced by presenting *swh1* in either DAT or NOM case. These experiments and their results are described in the following section. Our results show that, in German, (i) short elliptic sources exist and that (ii) varying the shiftiness of the embedded clause's subject *S_{emb}* has no impact on the acceptability of long source caMS configurations. We outline two possible analyses for this result and

declare our preference for the ‘short source only’ analysis, which states that the CC is not suspended in the presence of a stable S_{emb} , contrary to B&F (2022). According to our favored analysis, examples such as (8) are judged as acceptable because a short source interpretation is available for them.

Recall that the ‘short source only’ analysis appears unable to explain why (11), for which a short source interpretation is allegedly unavailable, is judged as acceptable. We discuss (11) and similar examples from B&F in Section 3, in which we present informally collected judgment data that suggest that a short source is available for (11) (and similar configurations from B&F) after all and that English caMS configurations in which a short source interpretation is genuinely precluded (or is at least rendered highly incongruous in the given context) receive lower acceptability judgments than expected if the CC was suspended in the presence of a stable S_{emb} .

2. Experiment on German MS

2.1. Design, predictions, participants

Design: We developed an acceptability judgment task with a 2×2 within-item design, where the two independent variables were SHIFTINESS (with levels *stable* and *shifty*) and SOURCE (with levels *full* and *short*). SHIFTINESS refers to whether the subject of the embedded clause co-refers with a DP in the matrix clause (*stable*) or not (*shifty*). SOURCE refers to whether a full or short source interpretation of the elliptic clause is being forced. Sample items exemplifying the four experimental conditions are presented in (12) and (13).

(12) Stable conditions

Nadine hat [einem Investor]_i erzählt, dass **er**_i Tanja einem Verkäufer nahelegen
 Nadine has a.DAT investor told that he.NOM Tanja a.DAT clerk recommend
 soll, aber ich weiß nicht genau,
 should but I know not exactly

‘Nadine told [an investor]_i that he_i should recommend Tanja to a clerk, but I don’t know...’

- a. welchem Investor welchem Verkäufer [full source]
 which.DAT investor which.DAT clerk

Interpretation: ‘... [to which investor]_i Nadine said that he_i should recommend Tanja to which clerk.’

- b. welcher Investor welchem Verkäufer [short source]
 which.NOM investor which.DAT clerk

Interpretation: ‘... which investor should recommend Tanja to which clerk.’

(13) Shifty conditions

Nadine hat [einem Investor]_i erzählt, dass **Tanja** ihn_i einem Verkäufer nahelegen
 Nadine has a.DAT investor told that Tanja him.ACC a.DAT clerk recommend
 soll, aber ich weiß nicht genau,
 should but I know not exactly

‘Nadine told [an investor]_i that Tanja should recommend him_i to a clerk, but I don’t know...’

- a. welchem Investor welchem Verkäufer [full source]
 which.DAT investor which.DAT clerk

Interpretation: ‘... [to which investor]_i Nadine said that he should recommend Tanja to which clerk.’

- b. welchen Investor welchem Verkäufer [short source]
 which.ACC investor which.DAT clerk

The experiment comprised 96 experiment items: 24 target items and 72 fillers. Fifteen fillers were *standard items* (Featherston 2009), which are known to elicit specific numerical scores and therefore function to prevent scale-distortion. Items were distributed across 4 lists, using a Latin square design. Participants were tasked with judging the acceptability of each experimental item on a 1-7 Likert scale, where 1 = *sehr schlecht* ‘very bad’ and 7 = *sehr gut* ‘very good’.

Participants: The experiment was conducted online using the *PsychoPy3* experiment creation software (Peirce et al. 2019), and participants were recruited via *Prolific* (www.prolific.co). Thirty-one

native speakers of German completed the experiment, but the results of three participants were excluded due to the participants’ misuse of the Likert scale (as revealed by the scores these participants provided for the standard filler items). Of the remaining 28 participants, 11 identified as female, 16 as male, and 1 as nonbinary. The mean age of participants was 31.4 (SD = 12.86).

Predictions: Our predictions are based on the assumptions that (i) B&F’s *shifty-subject* analysis of the CC is correct and (ii) short source interpretations of elliptic clauses are indeed available for caMS in which a DP in the matrix clause co-refers with a DP in the finite embedded clause, following Abels & Dayal (2017, 2022). We predict that:

- [P1] When a full source interpretation is forced, MS configurations exemplifying the *shifty* condition score significantly lower than similar configurations exemplifying the *stable* condition.
- [P2] When the short source interpretation is forced, there is no significant difference in ratings when the factor SHIFTINESS is varied.

2.2. Results

We obtained 168 individual ratings for each condition. The mean scores and standard deviation for each condition are presented in Table 1. Our statistical analysis involved fitting a cumulative link mixed model (CLMM) using the *ordinal* package (Christensen 2019) from the statistical software *R*, version 4.1.3. (R Core Team 2022). The model with the best fit was selected using a manual backward model selection process. This process involves taking a full model that includes all experimental factors and interactions as fixed effects and which also includes random intercepts and slopes for both subjects and items and then simplifying this full model, one factor at a time. The fit of each simplified model *m* is compared to the fit of its immediate, more complex predecessor *m*+1 using the *anova* function. (14) presents the formula for the model with the best structure supported by our data.

Table 1. Mean scores per condition			
SOURCE	SHIFTINESS	Mean	SD
<i>full</i>	<i>shifty</i>	3.69	1.54
<i>full</i>	<i>stable</i>	3.49	1.37
<i>short</i>	<i>shifty</i>	4.05	1.63
<i>short</i>	<i>stable</i>	4.13	1.66

(14) rating ~ source + (source * shiftiness | subject) + (1 | item)

Note that this formula does **not** include an interaction (*source * shiftiness*) or the combined main effect (*source + shiftiness*) term for the fixed effects. This is because the more complex model that modeled this interaction was not a significantly better fit for our results. In other words, no interaction between SOURCE and SHIFTINESS was observed in the results of our dataset. The model reports only a main effect for SOURCE, whereby test utterances in which the short source interpretation is forced are judged as significantly more acceptable than those test utterances in which the full source interpretation is forced (*z* = 2.546, *p* = 0.01, SE = 0.44), which confirms [P1]. The fact that no main effect was observed for SHIFTINESS disconfirms [P2].

2.3. Discussion

The observation that short source caMS is significantly more acceptable than full source caMS proves that short source caMS exists, which supports Abel & Dayal’s (2017, 2022) claims. Conversely, the observation that acceptability judgments remain constant when SHIFTINESS is modulated in the *full* condition shows that B&F’s *shifty-subject* analysis of the CC is not cross-linguistically viable.

But what is the cause of the main effect of SOURCE? Below, we outline two possible answers to this question. Before doing this, we must first introduce an independent result on German MS reported by Cortés Rodríguez (2022a:435-436). Although his study focused on islandhood and prototypical CC violations in German MS, part of his experimental design involved caMS configurations exemplified by

(15), in which the correlates are separated by a finite clause boundary and S_{emb} is shifty. Such utterances received a mean rating of 3.37 (SD = 1.71, number of participants = 29), which is lower than the comparable full source caMS from the current study (3.37 versus 3.69).

- (15) **Jeder** berichtete, dass Simon **an etwas** gedacht hat, aber ich weiß nicht
everyone reported that Simon about something thought has but I know not
wer an was.
who about what
'Everyone reported that S has been thinking about something, but I don't know who about what.'

2.3.1. Analysis-1: The CC and shiftiness are entirely unrelated

This analysis states that the main effect of SOURCE observed in our experimental results is due to a violation of the CC. This implies that the CC is not suspended when a stable item is present in the embedded clause of the antecedent in a caMS configuration. This analysis views the difference in ratings between (15) and (12a-13a) as entirely uninformative, i.e., an artifact of the fact that these judgments were obtained in two distinct experiments with slightly different experimental designs, targets, and fillers.

2.3.2. Analysis-2: The main effect of SOURCE reflects the complexity of the antecedent

This analysis states that the main effect of SOURCE observed in our results is due to the difference in the structural complexity of the antecedent between the *full* and *short* conditions. This analysis assumes that the more structurally complex an antecedent (in terms of hierarchical depth), the greater the processing burden and thus the lower the acceptability judgment, following e.g., Wasow & Arnold (2003). This analysis, therefore, claims that the main effect of SOURCE is unrelated to the CC.

Assuming momentarily that this analysis is correct, one must still explain why (15) and the full source caMS utterances from our current study exemplified by (12a-13a) receive different mean ratings. Two options present themselves:

Analysis-2a: Morphological case-marking obviates the CC. The first option is to treat the difference in ratings between (15) and (12a-13a) as uninformative, i.e., an artifact of the fact that these judgments were obtained in two distinct experiments with slightly different experimental designs, target items, and fillers. If true, then the CC is obviated in both (15) and (12a-13a), entailing that the CC simply does not apply in German in the first place. But why not? Assuming that B&F's shifty-subject analysis of English caMS is correct, one possibility is that the presence of morphological case-marking on sluiced wh-phrases somehow suspends the CC entirely. Thus, the CC is inoperative in German because this language morphologically case-marks its wh-phrases, but operative in English, which does not morphologically case-mark its wh-phrases. Problematically, this possibility seems rather implausible given that the CC is reported to be operative in a wide variety of other languages with morphologically case-marked sluiced wh-phrases; see Abels & Dayal (2022:7) for details.

Analysis-2b: The obviating effect of shiftiness on the CC is relativized across languages. The second option is to assume that the difference in ratings between (15) and (12a-13a) is informative. Because the main difference between (15) and (12a-13a) is the presence of a bound object pronoun in the embedded clause of the latter, it seems plausible to link the presence of the bound object pronoun with higher acceptability judgments. In other words, one could extend B&F's original proposal and postulate that languages differ regarding which stable items (if any) in the embedded clause of a caMS configuration will mitigate the negative impact on the acceptability of violating the CC. In English, only the presence of a stable referent in the subject position of the embedded clause provides this mitigative effect, whereas, in German, the presence of a stable referent in either the subject **or** the object position provides the mitigative effect. In essence, this proposal suggests that bound object pronouns/epithets are somehow 'inactive' for or 'invisible' to MS-related processes in English, whereas they are active and can therefore affect the acceptability of MS in German. Support for this proposal comes from the independent observation that bound object pronouns make for suitable correlates for MS in German but not English; compare (13a) to (16).

- (16) * Some student_i claimed that Mary introduced *him_i* to *some professor*, but I don't know which student ~~Mary introduced~~ to which professor. (adapted from Barros & Frank 2022:7)

It must be emphasized that our results are compatible only with an extended version of B&F's basic **empirical** claim, namely that the presence of a stable referent in the embedded clause of caMS configurations mitigates the negative impact on the acceptability of violating the CC. Our results are not aligned with B&F's **conceptual** claims, however. Recall that B&F's analysis (i) treats *swH2* as undergoing QR into the root clause, (ii) views QR as clause-bound for syntactic reasons, (iii) views the CC as a reflex of the syntactic locality restrictions on QR, and (iv) proposes that the syntactic locality restrictions on QR (and hence on the CC) are suspended when the clause from which the QRed phrase originates contains a stable subject. Firstly, our results do not support B&F's conceptual assumption that CC-violating caMS configurations are ungrammatical configurations. If one treats (15) as instantiating a canonical CC-violating configuration, then violating the CC in German MS yields a ~1.5-point decrease in acceptability, from 4.7 to 3.4. Because a score of 3.4 is higher than scores for the ungrammatical standard items D (2.99, SD = 1.43) and E (2.16, SD = 1.38) yet lower than scores for the grammatical standard item C (4.32, SD = 1.53), one cannot assume with any certainty that CC-violating caMS configurations in German are ungrammatical. Secondly, our results speak against B&F's claim that stable subjects play a special role in suspending the purportedly syntactic locality restrictions on QR (assuming the CC is a reflex of QR, of course). If analysis-2b is correct, then our results instead point towards a processing-based explanation of the CC, according to which the presence of a stable referent in the embedded clause reduces the processing burden of interpreting full source caMS. Under this analysis, the question of why English and German differ regarding which stable referents reduce the processing burden remains open, however.

2.3.3. Our preferred analysis is Analysis-1

Of the analytical options presented in Sections 2.2.1 and 2.2.2, we currently favor analysis-1, namely that the main effect of *SOURCE* is caused by the CC being violated in every German full source caMS configuration tested, regardless of whether the embedded subject was stable or shift. If B&F are correct that the CC is a reflex of illicit QR, then analysis-1 is fully compatible with the 3.37 and 3.69 mean ratings obtained in Cortés Rodríguez (2022a) and the current study for full source caMS, as QR over a finite clause boundary is known to produce only a mild-to-moderate degradation in acceptability judgments. For instance, Tanaka (2015) reports that nonelliptic utterances displaying QR over a finite clause boundary are rated as 3.4 on a 1-5 Likert scale, which is significantly worse than QR over no clausal boundary (rating = 4.0) but significantly better than QR from a complex NP island (rating = 2.4). Similar results are reported in Syrett & Lidz (2011) and Syrett (2015); see Wurmbrand (2018) for discussion.

If analysis-1 is correct, then the presence of a stable subject in the embedded clause does not suspend the CC in German caMS. This suggests that English caMS configurations such as (8) are not judged as acceptable because the CC is suspended but because hearers are basing their judgments on the short source interpretation, which is readily available (as our results show). Recall, however, that B&F counter this 'short source only' analysis of English caMS by presenting acceptable examples for which no short source interpretation is allegedly available. In the next section, we focus our attention on these examples. If these examples indeed (i) have no short source and (ii) receive an acceptability rating equivalent to standard, acceptable MS configurations in English, then the examples remain problematic for analysis-1 (i.e., the 'short source only' analysis). If neither (i) nor (ii) obtain, then analysis-1 receives additional support.⁵

⁵ In the interest of transparency, we must mention that Cortés Rodríguez (2022a:10) also obtains judgments for German caMS configurations exemplified by (i), in which the correlates are clausemates. Such utterances received a mean rating of 4.76 (SD = 1.61).

- (i) Simon berichtete, dass **jeder an etwas** gedacht hat, aber ich weiß nicht wer an
Simon reported that everyone about something thought has but I know not who about
was.
what
'Simon reported that everyone has been thinking about something, but I don't know who about what.'

Utterances such as (i) were therefore rated higher than the comparable short source caMS from the current study (4.76 versus ~4.1). Assuming that an increase in processing burden translates to lower acceptability judgments (see e.g., Fanselow 2021 and references in there), an explanation for why (i) was rated higher than (12b-13b) is readily available: (12b-13b) both involve associating *swH1* with a pronoun, which is an atypical correlate for sluicing. A processing cost is incurred here because the hearer is additionally burdened with the task of determining by what

3. Returning to apparent CC-obviations in English caMS

B&F claim that a short elliptic source is unavailable for each of the examples in (17) to (19). Because each example displays a stable embedded subject (i.e., a phrase that introduces no new d-ref) and is judged as acceptable, B&F offer these examples as evidence for their shifty-subject analysis of the CC.

Contrary to B&F, we believe that a short source interpretation **is** available for (17) to (19). These short sources are presented in the b-examples in (17) to (19). Not only is each short elliptic clause formally recoverable according to recent theories of syntactic identity under sluicing (Rudin 2019, Anand et al. 2022), but each short source interpretation is sensible in the discourse context, providing a different meaning to the full source interpretation in each case, yet a congruent one.

- (17)

Expletive ‘there’ as a stable embedded subject (B&F 2022:10, repeated from (11))

Some student claimed that there was a problem with some professor, but I can’t recall which student with which professor.

a.

... which student ~~claimed that there was a problem~~ with which professor.

[full source]

b.

... which student ~~had a problem~~ with which professor.

[short source]
- (18)

Negatively quantified DPs as a stable embedded subject (B&F 2022:10)

Some student lamented that no professor talked about a certain topic, but I can’t recall which student about which topic.

a.

... which student ~~lamented that no professor talked~~ about which topic.

[full]

b.

... which student ~~lamented~~ about which topic.

[short]
- (19)

Narrow-scoping existentially quantified DP as a stable embedded subject (B&F 2022: fn. 9)

Some linguist said that somebody might have written a paper about a Balkan language, but I can’t recall which linguist about which Balkan language.

a.

... which linguist ~~said that somebody might have written a paper~~ about which BL.

[full]

b.

... which linguist ~~wrote a paper~~ about which BL.

[short]

To begin exploring whether B&F’s native speaker consultants were providing acceptability judgments of the full source or the short source interpretations of (17) to (19), we collected judgments from 8 native speakers of English (6 linguists, including this paper’s second author; 2 non-linguists) not only on the examples in (17) to (19), but also on the examples in (20) to (22). The latter triplet of examples minimally differs from the former triplet in that the short source interpretation is highly incongruous in the specific context provided. Consultants were asked to provide a numerical rating on a 1-7 Likert scale (1 = unacceptable, 7 = fully acceptable). For (17), (19), and (21), we also presented the full and short sources (in a paraphrased form) and asked which interpretation(s) were available.⁶ These judgments were collected informally via email.

- (20)

Alternative to (17) (‘there’ existential)

Some student claimed that there was a pile of books in some professor’s office, but I can’t recall which student in which professor’s office.

a.

... which student ~~claimed that there was a POB~~ in which professor’s office.

[full]

b.

... which student ~~was~~ in which professor’s office.

[short]

type of DP the pronoun is bound. If bound by a DP that can take widest scope (e.g., *everyone, someone, a certain student*), then the pronoun can function as a suitable correlate for sluicing (Barker 2013). However, if the pronoun is bound by a DP that cannot take widest scope (e.g., *Sally, the boys*), then it cannot function as a suitable correlate (cf. B&F 2022:29).

⁶ There was a mistake in elicitation materials: we intended to present paraphrases of the full and short sources for (18) but actually presented paraphrases of the potential sources for its alternative in (21) instead. We believe that the high ratings attributed to (21) by some consultants are a direct repercussion of this mistake, with consultants revising their rating of (21) once they realized that the short interpretation, which would normally be irretrievable due to its high incongruity, is actually available. Two consultants explicitly mentioned this. For instance, consultant 4 said “when I first read [(21)], the [short source] interpretation didn’t occur to me. Now that it does, I’d say that only [the short source] is available, and is fully grammatical”.

- (21)

Alternative to (18) (negatively quantified DP)

Some student lamented that no professor spoke next to a certain lectern, but I can't recall which student next to which lectern.

a.

... which student ~~lamented that no professor spoke~~ next to which lectern.

[full]

b.

... which student ~~lamented~~ next to which lectern.

[short]
- (22)

Alternative to (19) (narrow-scoping 'somebody')

Some linguist said that somebody might have written a paper about a particular subatomic particle, but I can't recall which linguist about which subatomic particle.

a.

... which linguist ~~said that somebody might have written a paper~~ about which SP.

[full]

b.

... which linguist ~~wrote a paper~~ about which SP.

[short]

Table 2: Ratings for the English caMS examples in (17) to (22)									
Consultant no.		Linguists						Non-linguists	
		1*	2	3	4	5	6	7	8
E x a m p l e	(17) (there)	6	6	5	7	5	4	5	3
	full and/or short for (17)?	short	full	both	short	short	short	short	short
	(20) (there')	1	1	5	1	4	1	2	4
	(18) (no)	1	2	4	1	1	3	2	4
	(21) (no')	3	6	4	7	6	1	3	5
	full and/or short for (21)?	short	short	both	short	short	none	short	full
	(19) (somebody)	6	5	4	7	6	3	7	5
	full and/or short for (19)?	short	short	full	short	full	†both	both	short
	(22) (somebody')	1	1	3	1	1	3	6	6

* Consultant 1 is this paper's second author, † but with a preference for the short source

The fact that the results in Table 2 are highly varied is unsurprising, as the acceptability judgments assigned to English MS are known to be subject to a high degree of interspeaker variation (Kotek & Barros 2018, B&F 2022), especially when compared to German (Cortés Rodríguez 2022b). Despite this variability, some trends emerge. For most speakers, only a short source interpretation is available for B&F's example (17), with its alternative in (20), which only has a sensible full source interpretation, being consistently rated lower. For most speakers, B&F's example (18) is less acceptable than its alternative in (21) (once the incongruous short source interpretation of (21) was mistakenly pointed out to speakers, see footnote 7). The results for the 'somebody' cases in (19) and (22) are unclear, however.

These results are not predicted by B&F's analysis, which predicts that all native speakers will return results similar to consultant 3's.⁷ In other words, B&F expect that speakers will assign similar judgments to (17) to (19) and their corresponding alternatives in (20) to (22), and that speakers show a preference for a full source interpretation whenever the short source interpretation is incongruous. This expectation was not met in our small, informal study.

Of course, one cannot draw strong conclusions from these data. However, if a proper experimental study replicates the trends observed in Table 2, then this would provide convincing evidence that the CC is never obviated in caMS in languages such as English and German, and that the availability of short source interpretations is responsible for all apparent obviation of the CC.

4. Conclusion

We conducted an acceptability judgment experiment on German complex antecedent multiple sluicing (caMS) configurations in which a full or short source interpretation was forced through manipulating the morphological case-marking of the sluiced wh-phrases. The results show that full source caMS is significantly less acceptable than short source caMS, regardless of whether the subject of the

⁷ Consultants 3 and 6 are speakers of General American; the remaining consultants speak a British English variety. It appears that for consultant 3, the CC is indeed suspended in all cases tested. More information is needed to determine whether the CC is suspended in all or only stable subject caMS configurations for consultant 3, however.

embedded clause co-refers with an expression in the matrix clause (*stable*) or not (*shifty*). Our favored interpretation of these results is that (i) short source caMS exists (following Abels & Dayal 2017, 2022) and (ii) full source caMS always violates the Clausemate Condition (CC), which incurs an acceptability penalty. This latter claim stands in opposition to Barros & Frank (2022), who propose that the CC is suspended in full source caMS with a stable embedded clause subject. In Section 3, we presented informally collected judgments from 8 native speakers of English that cast serious doubts on the validity of the crucial ‘full source only’ English caMS data supporting Barros & Frank’s analysis.

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